Jyotish Kumar

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Summary

Systems Software and Backend Engineer with a foundation in **Mathematics and Computing** from **IIT Guwahati**. Builds **low-latency, multithreaded systems** in **C++** from first principles – a hand-rolled **epoll** event loop powering a matching engine load-tested to **64,500+ concurrent TCP connections**, and a custom key-value store sustaining **218,000+ RPS**. Validates work rigorously with **ASan/UBSan/TSan**, chaos, and endurance testing rather than treating a feature as done once it compiles.

Education

Indian Institute of Technology (IIT) Guwahati — Guwahati, Assam
Bachelor of Technology in Mathematics and Computing — *2021 – 2025*

- **Relevant Coursework:** Data Structures and Algorithms, Design and Analysis of Algorithms, Operating Systems, Database Management Systems, Computer Networks, Discrete Mathematics, Probability Theory, Linear Algebra.

Technical Skills

Languages: C++17, C, Python, JavaScript, SQL, HTML/CSS

Backend & Systems: epoll, Multithreading, Socket Programming, TCP/IP, Concurrency (mutexes, `shared_mutex`), Low-Latency Optimization, AOF Persistence, Non-Blocking I/O

Tools & Testing: Git, Linux, CMake, GitHub Actions (CI), GTest, Google Benchmark, AddressSanitizer, UndefinedBehaviorSanitizer, ThreadSanitizer, Bash

Core CS Knowledge: Data Structures, Algorithms, Operating Systems, Database Management, Computer Networks

Technical Projects

C++ Matching Engine | *C++17, epoll, TCP, Multithreading* — GitHub

- Built a single-instrument, price-time priority limit order book and TCP matching engine from scratch in C++17, driven by a hand-rolled single-threaded, edge-triggered, non-blocking **epoll** event loop with no external networking or matching libraries.
- Designed O(1) average-case order cancellation via an order-id-to-list-iterator index, and self-trade prevention logic so an order never matches against the same user's own resting orders.
- Benchmarked ∼**3.2M** non-crossing orders/sec in-process and **37–50μs** median / **54–160μs** p99 end-to-end latency over real TCP, flat from a 100-order to a **10-million-order** book; diagnosed a tail-latency regression traced to two unbounded query endpoints.
- Load-tested to **64,500** concurrent live TCP connections with zero failures (∼164k req/s aggregate); verified correctness with **105** unit/integration tests, all passing under AddressSanitizer+UBSan and ThreadSanitizer, plus chaos and fuzz testing (byte-level drip-feed, split reads, pipelined and malformed requests).

C++ Key-Value Store | *C++, Multithreading, TCP, AOF, GTest, GBench* — GitHub

- Engineered a high-concurrency, single-node key-value storage engine over raw TCP, implementing an **Append-Only File (AOF)** logging mechanism for reliable crash recovery.
- Architected a resilient network layer handling severe TCP stream chaos – byte-stream fragmentation (drip feed) and sticky batch writes – without buffer overflows.
- Scaled to **64,500** concurrent TCP client connections with zero failures (∼195k req/s aggregate); separately achieved a peak throughput of **218,251 requests/second** under 100 concurrent threads.
- Sustained **108.6 million** requests over a 10-minute endurance soak test with zero failures (∼181k RPS average), maintaining **25–50μs** median / **44–89μs** p99 latency across mixed SET/GET/DEL workloads from 100 to 1M live keys.

Algorithmic Achievements

- **LeetCode** (profile): Solved **751** problems with a peak contest rating of **1711** (Top 13.51%) and a 177-day active maximum streak.
- **GeeksforGeeks** (profile): Solved **603** problems, coding score of **2001**, a 198-day POTD streak, and an Institute Rank of 65.

Professional Experience

OCFP Faculty, K12 Techno Services — *May 2025 – Aug 2026*

- Taught Mathematics and Science, including foundational Mathematics for JEE and competitive exam preparation, and Physics for Grades 6-12.